THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The structure that gives the plant cell its definite shape is the: 1

(a) nucleus **(b)** cell wall **(c)** vacuole **(d)** cell membrane

2. The reflex action of removing the hand from a hot object involves: 1

(a) muscle only
(b) blood only
(c) bone only
(d) sensory neuron, relay neuron and motor neuron

3. A scientist finds an organism that has no cell wall, is multicellular, has a body with tissues, and feeds on other organisms. It is a(n): 1

(a) plant **(b)** fungus **(c)** protist **(d)** animal

4. Seeds dispersed by wind have: 1

(a) wings or hairs **(b)** fleshy covering **(c)** no coat **(d)** hooks

5. The troposphere is warmer near the ground mainly because: 1

(a) the Sun is closer
(b) greenhouse gases are absent
(c) ozone absorbs UV
(d) the ground is heated by sunlight and warms the air

6. Assertion (A): Plant cells do not burst when placed in distilled water. Reason (R): The cell wall resists excessive expansion of the cell. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A plant tissue is stained with safranin and some cells turn red. They contain: 1

(a) lignin **(b)** protein **(c)** starch **(d)** cellulose only

8. Dolphins and sharks look similar but belong to different classes because: 1

(a) dolphins have lungs, hair and mammary glands
(b) sharks have lungs
(c) dolphins have gills
(d) both have fins

9. The female reproductive structure that is the site of fertilisation and passage of the egg is the: 1

(a) oviduct **(b)** cervix **(c)** vagina **(d)** uterus

10. Assertion (A): Melting of polar ice accelerates global warming. Reason (R): Water has a lower albedo than ice. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Two cells A and B are placed in the same solution. A shrinks and B swells. This shows that the solution is: 1

(a) hypertonic to A and hypotonic to B **(b)** isotonic to both
(c) hypotonic to both **(d)** hypertonic to both

12. Assertion (A): The trunk of a tree becomes thicker every year. Reason (R): Cambium divides and adds secondary xylem and phloem. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

13. Assertion (A): Bryophytes are called the amphibians of the plant kingdom. Reason (R): They need water for the fertilisation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. Assertion (A): Cross-pollination leads to variation. Reason (R): It involves the fusion of gametes from different plants. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. If the albedo of the Earth decreases, the Earth's temperature will: 1

(a) increase **(b)** remain the same **(c)** become zero **(d)** decrease

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. What is the function of the cell membrane? Why is it called selectively permeable? 2

17. State two differences between the movement of materials in xylem and phloem. 2

18. Distinguish between homeothermic and poikilothermic animals with the classes. 2

19. Explain the role of the pollen grains and the ovule in seed formation. 2

20. State why India is vulnerable to climate change. Give two reasons. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Compare the life cycles of a moss and a fern briefly, noting the dominant stage in each. 3

22. A young plant grown from a seed shows features different from the parent while a plant grown from a stem cutting does not. Explain. 3

23. Why do deserts have large day-night temperature differences? Use the ideas of albedo and water vapour. 3

24. Describe how an artificial cell membrane (dialysis tubing) is used to demonstrate osmosis with glucose and starch. 3

OR

State the role of the vascular cambium and the cork cambium in a woody stem with their products. 3

25. State why some classification groups, such as algae, are not considered natural groups by modern taxonomists. 3

OR

Explain the different methods adopted by plants to prevent self-pollination. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A student models the Earth's temperature by the balance of incoming and outgoing energy. The Sun delivers 342 W/m squared on average, the albedo is 0.3. Calculate the absorbed energy, the outgoing energy needed at equilibrium and state what happens if the albedo decreases to 0.28 and the effect of more greenhouse gases. 5

OR

Explain cell specialisation and the differentiation of stem cells with examples from plants and animals, and discuss medical uses and ethical issues. 5

27. Explain the importance of structural adaptations of tissues for survival of plants in different habitats, with examples from xerophytes, hydrophytes and mesophytes. 5

OR

Explain the adaptations of plants to life on land from algae to angiosperms, listing the structures that appeared: roots, vascular tissue, seeds, flowers, fruits and their advantages. 5

END OF QUESTION PAPER — Total: 50 Marks

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